Ilari Angervuori

Researcher Skilled at Stochastic Modeling of Satellite Systems

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Github, *Google Scholar*, *LinkedIn*

Profile

A mathematically highly skilled R&D engineer in satellite communications with a pioneering publication record in stochastic geometry satellite system modeling, including a first-author publication in the high-impact journal Institute of Electrical and Electronics Engineers (IEEE) Transactions on Communications (TCOM) and a first-author presentation in the flagship conference IEEE GLOBECOM. A creative mindset that seeks simplicity, not complexity, and who is not afraid of fiddling with the details. Committed to contributing to the growing industry of satellite systems and to bringing in my unique expertise in a collaborative, challenging, development-driven environment. Always striving to grow professionally as well as spiritually, and to cultivate compassion and humility in all aspects of life. FI/EU citizen.

Skills and Expertise

• Data Analysis and Visualization • Inter-institute collaboration • Research • Stakeholder Management • Writing
• 3GPP NR NTN Specifications • C, Matlab, Python • Linux Operating System • Statistical Signal Processing

Experience and Impact

5/2025- **Senior Researcher**, *Aalto University*, Department of Electrical Engineering, Espoo, Finland
• Analyzed and simulated successive interference cancellation (SIC) algorithms that improved coverage performance by 900% for the third strongest Earth transmitter in terms of signal-to-interference ratio (SIR).
• Formulated a closed-form Gaussian autocorrelation function for the aggregate physical layer interference from Poisson-distributed signal terminals on the Earth surface, applicable, *e.g.*, to improving the efficiency of machine learning (ML) algorithms.
• Managed 2 R&D projects as the corresponding author and handled the full delivery pipeline (simulation, analysis, documentation, peer review) to publication.
• Mentored junior researchers as a bachelor's degree supervisor, establishing future talents.
• Peer reviewer in various IEEE journals, including IEEE TCOM, maintaining the quality of the leading journals in LEO satellite communications.

8/2023-2/2024 **Research Internship**, *University of Notre Dame*, Dept. of Electrical Engineering, South Bend, IN, US
• Based on rigorous mathematical analysis, a novel statistical SIR model was proposed by characterizing the SIR at a LEO satellite receiver using a Lomax distribution, providing important system-level insights into LEO network design.
• Coordinated a distributed team in a cross-functional project and handled the full delivery pipeline (simulation, analysis, documentation, peer review) to publication in IEEE TCOM.
• Identified an optimal cell size in a LEO uplink that maximizes average throughput and quantified a trade-off between average throughput and performance consistency. The results have clear implications for LEO network design.
• Secured project financing of 4349 € for personal internship expenses from the Väisälä Foundation.

05/2019-05/2025 **PhD Researcher**, *Aalto University*, Signal Processing of Wireless Networks, Espoo, Finland
• Built, maintained, and documented a multi-language simulation stack (GIT, MATLAB, Mathematica, Octave, Python) including algorithms for analysis, simulation, and visualization, used across several research projects.
• Conceived and executed an R&D project plan, ensuring the efficiency and impact of research.
• Developed a novel and well-cited stochastic geometry framework for the analysis of LEO networks, reducing simulation time by an order of magnitude 100 times more computationally efficient than the MATLAB Satellite Communication Toolbox, while providing equivalent results on the network performance metrics.
• Industry presentation at IEEE GLOBECOM involving a novel stochastic geometry system model for LEO uplink performance metrics, providing insight and tractable tools for LEO network design.

Education

Expected 2026 **Doctor of Science, PhD (Thesis in pre-examination.)**, *Aalto University, Dept. of Electrical Engineering*, Signal Processing for Communications, • Stochastic Geometry Analysis and Simulation of LEO networks

MSc, *Univ. of Helsinki*, Applied Analysis, • Finite Element Method (FEM) • Antenna Theory

BSc, *Univ. of Helsinki*, Applied Analysis, • Optimal Control Theory • System Performance Optimization

References

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○ Prof. Risto Wichman, Aalto University, risto.wichman@aalto.fi, +358 40 0800801
○ MSc. Abid Afridi, Aalto University, abid.afridi@aalto.fi